

Error Annotation for English Language Learning

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Abstract

We present a framework for annotating and evaluating English language learner (ELL) errors in written texts to support the development of educational tools that deliver targeted grammatical feedback. Two annotators labeled specific errors and assigned overall error categories using custom guidelines, with each annotating a separate subset of 368 CEFR-balanced samples. To assess consistency, we internally annotated 48 texts as a gold standard and computed inter-annotator agreement. Excluding surface-level issues, token-level agreement reached a micro-averaged F1 of 0.472, and document-level labels yielded a Cohen’s Kappa of 0.406. These results suggest moderate agreement and highlight the potential for developing automated grammar feedback tools.

1 Introduction

Our goal for this semester-long project was to train a model to identify English grammatical errors, particularly with the goal of helping those learning English as a second language increase their English proficiency. The model should not only be able to identify grammatical errors, but also what type of error is being made. In this way, learners can identify trends in their work and focus their attention on a specific area. It could, for example, inform a speaker that they are commonly making preposition errors, which may lead to the speaker reviewing preposition rules or paying more attention to their preposition usage. We also hope that our project can be a good basis for creating resources for L2 English learners.

1.1 Related Work

Our project builds on prior work in grammatical error annotation and learner corpora. [Han et al. \(2010\)](#) trained a model to correct preposition errors using Korean learner essays, showing that learner data yields better performance than native

corpora. [Rozovskaya and Roth \(2010\)](#) annotated multiple ELL error types across learners from nine L1 backgrounds and emphasized consistent annotation practices and full-sentence corrections. This approach informed our decision to annotate every error. [Lozano and Díaz-Negrillo \(2019\)](#) applied Interlanguage Annotation (ILA) to the COREFL corpus, analyzing morpheme-level accuracy and CEFR-related patterns. Their findings on persistent issues (e.g., third person singular -s) helped shape our error categories and proficiency-level expectations.

These studies highlighted the value of high-quality learner annotations for building correction tools and guided our decisions on error scope, annotation strategy, and evaluation.

2 Guidelines

We created guidelines which would aid annotators in identifying and categorizing errors in a given document, as well as categorizing the documents as a whole by proficiency level. We identified some common error types that are present in the data and created labels and detailed examples for each. In addition, we pre-annotated spelling, punctuation, and capitalization errors by running the data through a script, and clarified that Britishisms should not be categorized as errors.

Regarding proficiency labeling, annotators were advised on how to label documents as beginner, intermediate, or advanced based on criteria laid out in the guidelines. Over the course of writing the guidelines, we modified the Verb Tense and Subject/Expletive Omission tags to include a greater diversity of errors present in the data set. We also eliminated dependency usage for this task, as we found it unnecessary, too complicated, and unintuitive to use in the annotation software. Lastly, we added an annotation hierarchy to disambiguate what label a particular error should receive when

more than one tag could plausibly apply.

3 Dataset

The corpus we are using for this project is COREFL: Corpus of English as a Foreign Language (Lozano et al., 2020). It is derived from text generated by German and Spanish learners of English along with native control subcorpora. Each document is a response to one of a number of prompts, for example, 'Tell the story shown in the pictures. [...]'. Detailed metadata accompany each text file, covering aspects such as participant identifiers, placement test scores, CEFR-aligned proficiency levels, demographic and linguistic background information (including L1, parental languages, and language exposure), as well as task-specific variables such as task title, medium, time taken, and resources used. COREFL is distributed under the Creative Commons CC BY-NC-ND 3.0 ES license.

4 IAA and Adjudication

We had two external graduate student annotators provide annotations for our task. Each annotator was given 184 samples: 35 randomly-selected samples from each CEFR proficiency level A1-C1, and 9 randomly-selected samples from C2. The samples were randomized, and there was no overlap of data between external annotators.

Inter-annotator agreement (IAA) was assessed using a separate internally annotated evaluation set. Our internal team selected 24 samples from each annotator’s dataset, a total of 48 samples, and annotated them independently to serve as a gold standard. To focus on the grammatical and word choice error types, we excluded spelling, punctuation, and capitalization errors from the evaluation.

We evaluated each annotator’s performance against the gold standard using both token-level and document-level metrics. At the token level, Annotator 1 achieved higher performance (micro F1 = 0.566) than Annotator 2 (micro F1 = 0.289), with overall token-level scores falling between them (micro F1 = 0.472). Macro-averaged scores were generally lower, reflecting class imbalance and greater difficulty with less frequent error types. At the document level, overall accuracy was 0.604 with a Cohen’s Kappa of 0.406, indicating moderate agreement. Annotator 1 achieved a higher document-level accuracy score than Annotator 2, with scores of 0.708 and 0.500 respectively. Annotator 1 also

exhibited higher consistency with the gold labels ($\kappa = 0.562$ vs. 0.250). Full precision, recall, and F1 scores are provided in Tables 1, 2, and 3.

Level	Metric	Annotator 1	Annotator 2	Overall
Token-level (Micro)	Precision	0.561	0.275	0.462
	Recall	0.571	0.303	0.483
	F1 Score	0.566	0.289	0.472
Token-level (Macro)	Precision	0.472	0.238	0.355
	Recall	0.486	0.307	0.396
	F1 Score	0.475	0.260	0.367
Document-level	Accuracy	0.708	0.500	0.604
	Cohen’s Kappa	0.562	0.250	0.406

Table 1: Comparison of token-level and document-level metrics across annotators and overall

4.1 IAA Result Evaluation

Token-level grammatical mistake annotation was more challenging than anticipated despite detailed labeling guidelines. As shown in table 2, errors like `article_error` (F1 = 0.585) and `verb_tense_error` (F1 = 0.568) showed reasonably strong performance, indicating annotator alignment on these relatively surface-level patterns. In contrast, more ambiguous or infrequent types like `pronoun_case_error` (F1 = 0.167), `subject-verb_order_error` (F1 = 0.000), and the label `other` (F1 = 0.275) exhibited substantially lower agreement.

We observed that annotators often disagreed on the precise span of the error. One annotator might mark only the problematic word, while another might highlight a full phrase. These span-level inconsistencies contributed to reduced token-level overlap even when annotators and the gold standard conceptually agreed on the nature of the error. This suggests that strict span-matching in evaluation may underestimate true annotation alignment.

Error Type	Precision	Recall	F1
<code>article_error</code>	0.533	0.649	0.585
<code>nominalization_error</code>	0.600	0.750	0.667
<code>number_agreement_error</code>	0.778	0.700	0.737
<code>other</code>	0.236	0.329	0.275
<code>preposition_error</code>	0.542	0.482	0.510
<code>pronoun_case_error</code>	0.143	0.200	0.167
<code>subject-verb_agreement_error</code>	0.522	0.474	0.497
<code>subject-verb_order_error</code>	0.000	0.000	0.000
<code>subject/expletive_omission_error</code>	0.632	0.343	0.444
<code>verb_tense_error</code>	0.520	0.627	0.568

Table 2: Overall token-level performance by error type

At the document level, table 3 shows that the beginner and advanced texts were classified with higher reliability (F1 = 0.743 and 0.629, respectively), while intermediate-level texts proved more difficult (F1 = 0.385), suggesting less consistent linguistic cues or greater subjective variability in this range.

Annotator-level performance further illustrates this disparity. Annotator 1 achieved high scores across all three levels, with particularly strong performance on intermediate texts (F1 = 0.667), while Annotator 2 failed to identify any intermediate-level documents correctly (F1 = 0.000). This suggests not only a lack of agreement but a fundamental difference in how annotators interpreted the proficiency criteria, particularly for mid-range texts that lacked clear indicators. These findings underscore the need for more refined, transparent guidelines when conducting document-level CEFR annotation.

Proficiency Level	Precision	Recall	F1
Overall			
Advanced	0.688	0.579	0.629
Beginner	0.813	0.684	0.743
Intermediate	0.313	0.500	0.385
Annotator 1			
Advanced	0.625	1.000	0.769
Beginner	0.875	0.583	0.700
Intermediate	0.625	0.714	0.667
Annotator 2			
Advanced	0.750	0.429	0.546
Beginner	0.750	0.857	0.800
Intermediate	0.000	0.000	0.000

Table 3: Document-level performance by predicted proficiency level

5 LLM Annotation Approach

To evaluate the performance of a general-purpose large language model (LLM) on our learner error annotation task, we used the same set of 48 gold-standard annotated learner texts created for IAA analysis. Each document was input into the LLM via the OpenAI GPT-4o API (OpenAI et al., 2024) with a prompt based on our annotation schema, and the LLM was instructed to return both word-level error annotations and a single document-level CEFR proficiency label. Each document was annotated individually in a zero-shot setting. Errors were tagged based on the same predefined label set, and we compared the LLM’s output against the gold-standard annotations using the same evaluation metrics as our IAA evaluation.

5.1 LLM Annotation Evaluation

As shown in Table 4, the LLM consistently over-predicted "Beginner" labels, resulting in high recall but very low precision for that class. It struggled significantly with the "Intermediate" and "Advanced" categories. Overall accuracy was 0.438,

Level	Class/Metric	Precision	Recall	F1 Score
Document-level (CEFR)	Beginner	0.613	1.000	0.760
	Intermediate	0.063	0.100	0.077
	Advanced	1.000	0.053	0.100
	Overall Accuracy		0.438	
Token-level (All errors)	Micro Avg	0.027	0.020	0.023
	Macro Avg	0.017	0.020	0.018

Table 4: LLM performance on document-level CEFR classification and token-level error annotation.

with weak agreement beyond chance (Cohen’s Kappa = 0.156).

At the token level, the model produced extremely low scores (micro F1 = 0.023), frequently produced overlapping or imprecise spans and often labeled errors in different positions than the gold-standard, even when referring to the same general issue. This misalignment led to extremely low precision and recall across samples. This was especially common for more abstract error types such as `preposition_error` or `subject/expletive_omission_error`. Although its feedback was occasionally semantically valid, it did not meet the strict alignment criteria used in evaluation, thus resulting in near-zero F1 scores for most samples.

While the LLM did have basic ability to identify some of the errors in learner writing, it often failed to accurately identify the correct error types. Without fine-tuning or domain adaptation training, these zero-shot annotations are not sufficient for quality-grading grammatical error marking or stable CEFR classification. The human annotators were far more consistent with the gold standard, particularly for more nuanced grammatical categories. This comparison mirrors the current limitations of general-purpose LLMs in error annotation tasks and points to the need for human knowledge or model training to tasks for better learner language evaluation.

6 Error Analysis

Beyond agreement metrics, further qualitative analysis of annotation behavior revealed patterns of ambiguity, inconsistency, and subjectivity. These observations help explain underlying annotation difficulties and inform future improvements.

6.1 Labels Ambiguity

Some tags such as `preposition_error` and `subject/expletive_omission_error` were often the cause of conflict. For example, in cases such as "go to school" vs. "go the school," annotators sometimes disagreed as to whether the issue was preposition omission or article error.

The remaining tags like the `nominalization_error` were applied on occasion or otherwise. Annotators struggled with deciding whether ill-phrased expressions (such as "do the explanation") were a grammatical nominalization error or mere stylistic awkwardness. Annotators used the label "other" as a fall-back, reducing recall and precision in those categories.

6.2 Divergent Correction Strategies

Another source of disagreement stemmed from the flexibility of grammatical correction strategies. In certain cases, multiple valid corrections existed, and annotators made divergent, but still reasonable choices. For instance, some learner texts contained tense inconsistency: parts of a narrative might shift between present and past tense. One annotator might standardize all verbs to the past tense, while another might correct the same verbs into consistent present tense.

Similarly, errors involving pronouns occasionally led to disagreements. If a learner used an inconsistent pronoun (e.g., switching between "he" and "she"), annotators had to make a judgment about the intended referent. One annotator might standardize all references to "he," while another chose "she." While both are grammatically valid corrections, they reflect differing assumptions and thus affect word-level annotation overlap, leading to mismatches with the gold standard.

These cases highlight the inherent subjectivity in error correction and underscore the importance of flexible evaluation metrics or adjudication processes that account for variation in plausible revisions.

6.3 Refinement and Limitations

The quality of annotation was vulnerable to language background and individual interpretation, all of which affected IAA. For control, we also compared the performance of human annotations against zero-shot LLM-produced annotations. The LLM fared poorer than human annotators across all the metrics, with significantly low F1 scores both at token and document levels. While it detected surface-level errors, it lacked depth and subtlety for linguistic interpretation and repeatedly misclassified CEFR levels.

All in all, our gold-standard annotations provide a solid foundation for follow-up model development and testing, and the IAA analysis offers a

realistic benchmark against which to measure annotation difficulty and task complexity.

7 Future Directions

7.1 Expand the Dataset

We would like to expand our dataset to include more L1 languages. It would give us more insight on the types of errors that are prevalent in ELL data. We expect that we may have a more even distribution of labels used in our annotated data. There may also be additional error types we would end up including as labels, as data from speakers of different languages would offer different phenomena to explore.

7.2 Refine the Guidelines

We ran into some inconsistencies with Label Studio, and we would like to further refine our guidelines to accommodate for this. Some of our token- and span-level annotations included the space that preceded or followed the annotated text, and some did not. This occurred within both our externally-annotated data and our internally-annotated data. We did not include this specific detail in our guidelines, however we did require certain annotations on empty spaces. It may have skewed the results having this discrepancy in our annotated data.

7.3 Use Cases

We would be interested in exploring different use cases for our work, such as creating a system that could aid English language learners in identifying errors in their writing, or to create a baseline for research on improving educational materials related to teaching English as an L2 language.

8 Conclusion

This project evaluated token-level and document-level agreement between two annotators and a gold standard in the context of English learner error annotation. Our findings revealed moderate overall agreement. Certain error types, such as article and verb tense errors, were more reliably annotated than other error types, such as pronoun case or subject-verb order. These results highlight both the value of clear annotation guidelines and the challenges of subjective label boundaries. Future work will build on these insights to improve annotation consistency and system performance.

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